Number Of Ways To Make Change

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The problem

- Count combinations of unlimited coins that sum to n.

Intuition

- Count combinations by processing one coin at a time so order doesn't matter.

Approach

- 1. ways of zero is one
- 2. For each coin, for each amount, add ways of amount minus coin
- 3. Return ways of n

Java solution

```java
public class NumberOfWaysToMakeChange {
    public static int numberOfWaysToMakeChange(int n, int[] denoms) {
        int[] ways = new int[n + 1];
        ways[0] = 1;
        for (int d : denoms) for (int a = d; a <= n; a++) ways[a] += ways[a - d];
        return ways[n];
    }
}
```

Complexity

- Time: O(n*d)
- Space: O(n)

Recap

- Count combinations by processing one coin at a time so order doesn't matter.
- Complexity: O(n*d)